```
> restart:

  # Compute p(0),...,p(Nmax) by Euler's pentagonal recurrence.
  BuildPartitionTable := proc(Nmax::nonnegint)
      local Ptab, nn, r, g, sign, total;

      Ptab := Array(0..Nmax, datatype=anything, fill=0);
      Ptab[0] := 1;

      for nn from 1 to Nmax do
          total := 0;
          r := 1;
          sign := 1;

          while true do
              g := iquo(r*(3*r-1), 2);

              if g > nn then
                  break;
              end if;

              total := total + sign*Ptab[nn-g];

              if g+r <= nn then
                  total := total + sign*Ptab[nn-g-r];
              end if;

              r := r+1;
              sign := -sign;
          end do;

          Ptab[nn] := total;
      end do;

      return Ptab;
  end proc:

  # Compute N(u,nn) using (2.4) with iota=2.
  # Negative partition arguments contribute zero.
  # This formula also handles u=0 when nn>=1.
  RankCoeff := proc(u::nonnegint, nn::posint, Ptab)
      local r, x, term, total, sign;

      total := 0;
      r := 1;
      sign := 1;
```

```
    while true do
        x := nn-r*u-iquo(r*(3*r-1), 2);

        if x < 0 then
            break;
        end if;

        term := Ptab[x];

        if x >= r then
            term := term-Ptab[x-r];
        end if;

        total := total + sign*term;
        r := r+1;
        sign := -sign;
    end do;

    return total;
end proc:

# Exact number of required determinants with 73 <= n <= H.
Count33Through := proc(H::nonnegint)
    local q, rem3;

    if H < 73 then
        return 0;
    elif H <= 106 then
        return iquo((H-73)*(H-72), 2);
    end if;

    q := iquo(H-7, 3);
    rem3 := irem(H-7, 3);

    return iquo(3*q*(q-1), 2)+(rem3+1)*q-1056;
end proc:

# Verify every required determinant in the chosen n-block.
Verify33 := proc(nFirst::posint, nLast::posint)
    local Ptab, nn, m, bound, left, mid, right,
          det, checked, expected;

    if nFirst < 73 or nLast < nFirst or nLast >= 10^6 then
        error "require 73 <= nFirst <= nLast < 10^6";
    end if;
```

```
    expected := Count33Through(nLast)-Count33Through(nFirst-1);

    printf("Computing p(0),...,p(%a).\n", nLast);
    Ptab := BuildPartitionTable(nLast);
    printf("Partition table ready; %a determinants to check.\n",
           expected);

    checked := 0;

    for nn from nFirst to nLast do
        bound := min(nn-73, iquo(nn-7, 3));

        # An empty m-range needs no comparison.
        if bound >= 1 then
            left := RankCoeff(0, nn, Ptab);
            mid := RankCoeff(1, nn, Ptab);

            for m from 1 to bound do
                right := RankCoeff(m+1, nn, Ptab);
                det := mid^2-left*right;

                if det < 0 then
                    error "FAIL: n=%1, m=%2, determinant=%3",
                          nn, m, det;
                end if;

                checked := checked+1;

                # Reuse the adjacent coefficients.
                left := mid;
                mid := right;
            end do;
        end if;

        if irem(nn, 1000) = 0 or nn = nLast then
            printf("Completed n=%a; checked=%a.\n", nn,
checked);
        end if;
    end do;

    if checked <> expected then
        error "incomplete verification: checked=%1,
expected=%2",
              checked, expected;
    end if;
```

```
    printf("PASS: %a <= n <= %a; checked %a determinants.\n",
           nFirst, nLast, checked);

    return [nFirst, nLast, checked, true];
end proc:

# Full range in (3.3).
# Change only these two bounds to verify a smaller block.
nFirst := 50000:
nLast := 80000:

result := Verify33(nFirst, nLast):
Computing p(0),...,p(80000).
Partition table ready; 649941664 determinants to check.
Completed n=50000; checked=16664.
Completed n=51000; checked=16847497.
Completed n=52000; checked=34011664.
Completed n=53000; checked=51509164.
Completed n=54000; checked=69339997.
Completed n=55000; checked=87504164.
Completed n=56000; checked=106001664.
Completed n=57000; checked=124832497.
Completed n=58000; checked=143996664.
Completed n=59000; checked=163494164.
Completed n=60000; checked=183324997.
Completed n=61000; checked=203489164.
Completed n=62000; checked=223986664.
Completed n=63000; checked=244817497.
Completed n=64000; checked=265981664.
Completed n=65000; checked=287479164.
Completed n=66000; checked=309309997.
Completed n=67000; checked=331474164.
Completed n=68000; checked=353971664.
Completed n=69000; checked=376802497.
Completed n=70000; checked=399966664.
Completed n=71000; checked=423464164.
Completed n=72000; checked=447294997.
Completed n=73000; checked=471459164.
Completed n=74000; checked=495956664.
Completed n=75000; checked=520787497.
Completed n=76000; checked=545951664.
Completed n=77000; checked=571449164.
Completed n=78000; checked=597279997.
Completed n=79000; checked=623444164.
Completed n=80000; checked=649941664.
PASS: 50000 <= n <= 80000; checked 649941664 determinants.
>
```